

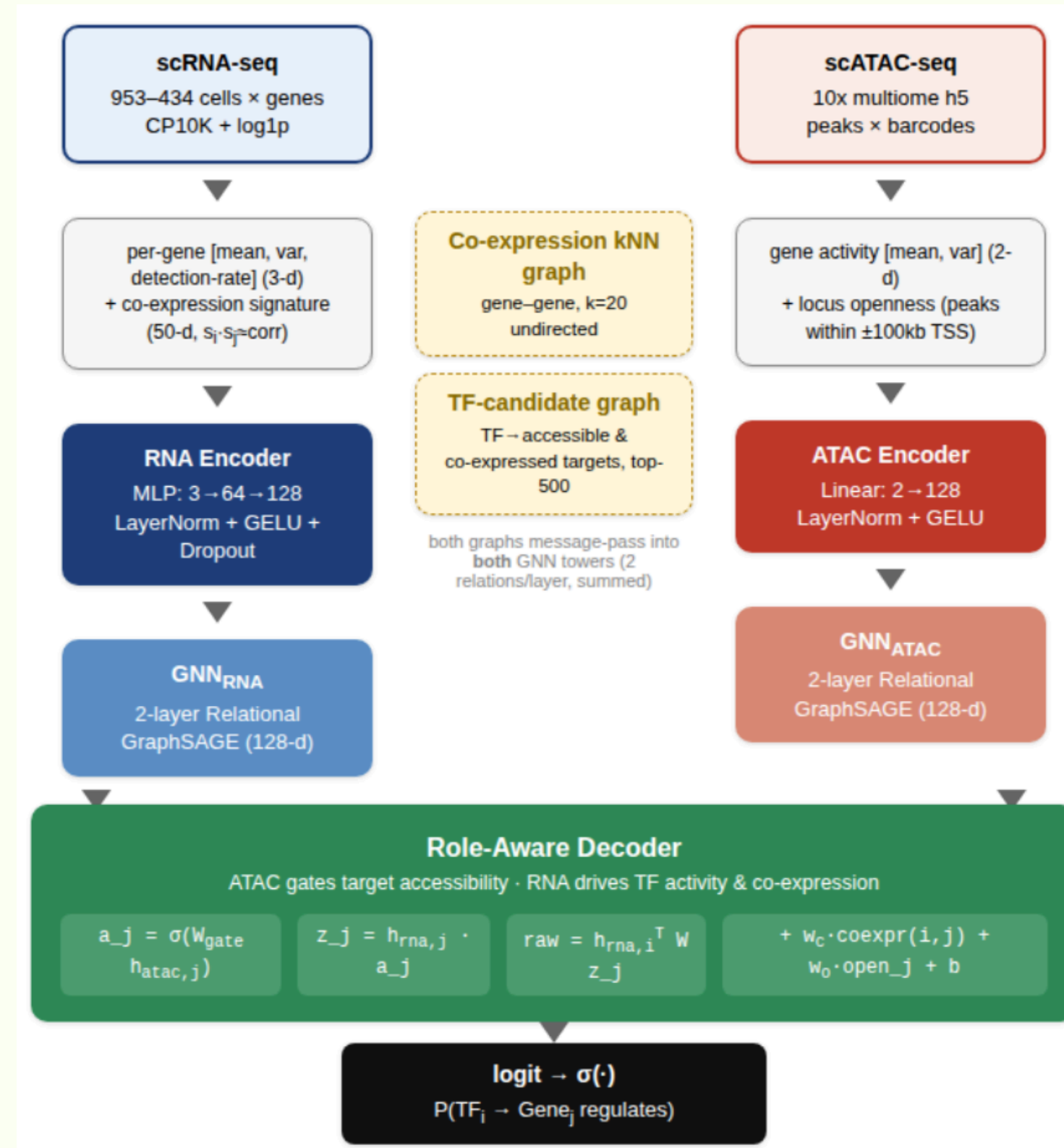
GENE REGULATORY NETWORK INFERENCE FROM INTEGRATIVE MULTI-OMICS DATA

By - Vishak Kashyap K, UG4 CND
Advisor - Dr Vinod PK

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Weekly Update

Updated Architecture



About the dataset

Metric	K562 (human)
scRNA cells	~950
scATAC cells	~1,100
Transcription factors (TFs)	209
Target genes	~22,900
Localization edges	~1.17 M
Perturbation edges	~173 K
Dual-evidence edges	~27 K

Results against baselines

Method	Loc	Pert	Dual
MEvD-GRN	0.420	0.575	0.558
scMultiomeGRN	0.831	0.523	0.504
GRNBoost2	0.535	0.206	0.173
RegDiffusion	0.546	0.185	0.165
GMF-GAE	0.526	0.218	0.160

Not a very significant improvement!

Ablations - No GNN

	Loc AUPR	Pert AUPR	Dual AUPR
full model	0.410	0.569	0.546
no GNN	0.498	0.341	0.241

- Key Takeaway: Eliminating graph convolutions causes the dual-evidence AUPR to drop drastically from 0.546 to 0.241, confirming that topological message passing provides essential predictive signal.
- Critical Context: While this confirms that graph smoothing is critical to the pipeline, it merely proves that relational information helps; it does not substantiate our specific multi-relational hypothesis. Subsequent ablations show this improvement is driven almost entirely by expression correlation alone.

Ablations - No ATACseq

	Loc AUPR	Pert AUPR	Dual AUPR
full model	0.410	0.569	0.546
RNA only	0.414	0.540	0.543

- Key Takeaway: Removing the entire scATAC-seq pipeline causes a negligible decline in dual-evidence AUPR from 0.546 to 0.543, indicating that chromatin accessibility contributes almost nothing to the model's final predictions.
- Critical Context: Because the model performs equally well without accessibility data, our "role-aware accessibility gate" is largely non-functional, and the openness filter underlying our TF-candidate prior relies on a largely uninformative signal.

Ablations - Dual Prior Graphs

Run	Loc AUPR	Pert AUPR	Dual AUPR
both graphs	0.404	0.507	0.451
coexpr only	0.418	0.441	0.383
TF-candidate only	0.418	0.374	0.258

- Key Takeaway: Message passing over the co-expression graph accounts for the vast majority of performance, whereas using the TF-candidate graph in isolation drops dual-evidence AUPR to 0.258 which is comparable to removing the GNN entirely (0.241).
- Critical Context: The "dual prior" regulatory design is largely an illusion; our GNN behaves almost entirely as a single-cell expression smoother rather than an integrative multi-relational reasoner.

Ablations Summary

Ablation	Dual AUPR	What it exposed
no_gnn	0.241	need message passing
tf_cand_only	0.258	TF-candidate prior is weak alone
coexpr_only	0.383	co-expression does most GNN work
both graphs	0.451	second prior adds little
shipped curriculum	0.546	competitive headline, broken loc
rna_only	0.543	ATAC not used
gated / concat fusion	0.546 / 0.555	role-aware decoder not the win
loc_only	0.663	binding signal transfers some, not enough

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- Critical Context: The "dual prior" regulatory design is largely an illusion; our GNN behaves almost entirely as a single-cell expression smoother rather than an integrative multi-relational reasoner.



THANK YOU!

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